Phase 4: Understanding the Complete BRIO Estimator

Detailed Theory, Worked Examples, Factor Graphs, and Optimization

Study Notes for Reproducing

A Robust Baro-Radar-Inertial Odometry M-Estimator for Multicopter Navigation in Cities and Forest

1 Purpose and Connection to Phases 1–3

Goal: Understand how the estimator uses initial values and multiple sensor constraints to obtain an optimized sequence of drone states. Phase 1 introduced the unknown state; Phase 2 introduced coordinate frames and sensor measurements; Phase 3 introduced sensor residual functions. Phase 4 connects these pieces.

Key Idea

Measurements $\rightarrow$ Factors and graph $\rightarrow$ Optimization $\rightarrow$ State estimates

A factor is a *function defining a measurement constraint*, not a final corrected measurement. The residual depends on candidate state values and is evaluated during optimization.

1.1 The State We Estimate

From Section IV-A, Eq. (3), the state at radar frame i is

$$x_i = (R^i_{IB},\ {}^{I}t^i_{IB},\ {}^{I}v^i_{IB},\ {}^{B}b^i_g,\ {}^{B}b^i_a). \tag{1}$$

It has 15 degrees of freedom: orientation (3), position (3), velocity (3), gyroscope bias (3), and accelerometer bias (3). $R_{IB} \in \mathrm{SO}(3)$ maps body-frame coordinates into inertial-frame coordinates. In the paper, I is the inertial frame, B the body/IMU frame, and R the radar frame.

1.2 The Main Question

At t_0 we initialize x_0. When the drone moves, IMU integration can predict a new state, but integrated IMU noise and biases lead to drift. Radar supplies radial-velocity constraints; barometry supplies altitude constraints; tracked zero-Doppler targets can help at standstill. Different sensors constrain *different aspects* of the state, so we estimate the states jointly.

2 Topic 9: Initial State and Prior Factor (Section IV-H, Eqs. 16–21)

2.1 Initialization Is Not the Same as a Prior Factor

Initialization supplies values to start the estimator; the prior factor penalizes disagreement between an estimated initial state and the fixed initial references.

State component	Initial reference specified in the paper
Position	Zero (origin of the local inertial frame)
Velocity	Zero (drone initially stationary)
Orientation	Orientation from a parallel Madgwick filter aligned with the attitude controller
Gyroscope bias	Average of the initial gyroscope messages, given a strong prior
Accelerometer bias	Zero

A tilde denotes the *initial reference*, e.g., $\widetilde{v}_0$. An untilded quantity such as v_0 denotes the optimizer's current candidate value. The reference stays fixed while the candidate may change.

2.2 Orientation: Roll, Pitch and Yaw

Orientation describes how the drone is rotated, not where it is located. The paper's body axes point $+x$ forward, $+y$ left, and $+z$ up.

Angle	Symbol	Rotation axis	Intuitive meaning
Roll	ϕ	Body x	Sideways tilt
Pitch	θ	Body y	Forward/backward tilt
Yaw	ψ	Body z	Change of horizontal heading

The paper stores orientation as a rotation matrix R_{IB} rather than directly as three angles. A rotation matrix allows the estimator to convert body-frame vectors to inertial-frame vectors.

2.3 Why Orientation Matters for IMU Integration

The IMU senses acceleration-related specific force in the *body frame*; the state stores velocity and position in the *inertial frame*. Thus orientation is necessary to interpret the measurements.

$$ {}^{I}u = R_{IB}\,{}^{B}u \qquad \text{(coordinate transformation of a vector)}, \tag{2} $$

$$ {}^{I}a_{\text{motion}} \approx R_{IB}({}^{B}f_{\text{meas}} - {}^{B}b_a) + {}^{I}g \qquad \text{(simplified IMU motion model)}. \tag{3} $$

Here f_{meas} is measured specific force, b_a accelerometer bias, and g the gravity acceleration vector in inertial coordinates. The exact implementation also models noise and changing orientation. Orientation errors become velocity and position errors after integration.

$$ \boxed{\text{Orientation} \rightarrow \text{Inertial acceleration} \rightarrow \text{Velocity} \rightarrow \text{Position}.} \tag{4} $$

2.4 What the Madgwick Filter Does

The Madgwick filter is a separate orientation estimator. In its common IMU-only form it: (1) integrates gyroscope angular velocity to predict orientation; (2) uses the accelerometer's gravity reference to correct roll and pitch; and (3) maintains orientation conveniently as a quaternion, which can be converted to R_{IB}. The standard method uses a gradient-descent correction to reduce reference-direction disagreement.

Key Idea

With only gyroscope and accelerometer measurements, *absolute yaw is not observable*: two stationary drones facing north and east can have the same gyro and gravity readings. Gyroscope integration tracks yaw *changes* but suffers drift. A heading reference would be necessary to identify absolute north/east. The paper says a parallel Madgwick filter provides alignment with the attitude controller; it does not specify all internal filter settings.

2.5 How Gyroscope Bias Is Initially Calibrated

At standstill, true rotation rate is approximately zero, but the gyroscope has an offset. Measurements arrive repeatedly over time, with three axes per timestamp. The initial bias estimate is approximately

$$\boxed{{}^{B}\widetilde{b}_{g} = \frac{1}{N}\sum_{n=1}^{N} {}^{B}\omega_{\text{meas},n}} \qquad \text{(stationary startup, per axis).} \tag{5}$$

Illustrative Example (Not Experimental Data)

Five *illustrative* sequential x-axis readings (rad/s) are $0.019, 0.021, 0.020, 0.022, 0.018$. Their average is 0.020 rad/s. These are five *timestamps*, not five sensors. During flight, use the *current* reading, e.g., $\omega_{\text{meas}} = 0.520$ rad/s, giving $\omega_{\text{corrected}} \approx 0.520 - 0.020 = 0.500$ rad/s. The same correction applies independently to x, y, z.

The paper does not state the exact number N of initial messages used for this calibration. Gyroscope bias is initialized with high certainty because yaw-rate bias is difficult to observe from the presented radar/IMU setup.

2.6 Prior Residuals: Eqs. (16)–(20)

Using paper-consistent notation, the residual components are

$$r_R^0 = \operatorname{Log}\left((R_{IB}^0)^{\mathsf{T}}\widetilde{R}_{IB}^0\right), \qquad \text{orientation, Eq. (16),} \tag{6}$$

$$r_p^0 = {}^{I}t_{IB}^0 - {}^{I}\widetilde{t}_{IB}^0, \qquad \text{position, Eq. (17),} \tag{7}$$

$$r_v^0 = {}^{I}v_{IB}^0 - {}^{I}\widetilde{v}_{IB}^0, \qquad \text{velocity, Eq. (18),} \tag{8}$$

$$r_{b_g}^0 = {}^{B}b_g^0 - {}^{B}\widetilde{b}_g^0, \qquad \text{gyroscope bias, Eq. (19),} \tag{9}$$

$$r_{b_a}^0 = {}^{B}b_a^0 - {}^{B}\widetilde{b}_a^0, \qquad \text{accelerometer bias, Eq. (20).} \tag{10}$$

Log maps the difference between two rotations into a 3-dimensional rotation error; it is not an ordinary subtraction of rotation matrices. If the current and reference rotations agree, $r_R^0 = 0$. The paper's printed Eq. (21) appears to repeat the position term in the stacked notation; logically the five residuals are orientation, *position*, *velocity*, gyroscope bias and accelerometer bias, consistent with Eqs. (16)–(20):

$$\boxed{r_P^0 = \begin{bmatrix} r_R^0 \\ r_p^0 \\ r_v^0 \\ r_{b_g}^0 \\ r_{b_a}^0 \end{bmatrix} \in \mathbb{R}^{15}.} \tag{11}$$

Illustrative Example (Not Experimental Data)

Suppose the reference initial position is $(0, 0, 0)$, but a current candidate is $(0.1, 0, 0)$ m. Then $r_p^0 = (0.1, 0, 0)$ m. If both are $(0, 0, 0)$, the residual is zero. This does not mean the estimator recomputes the initialization: it *tests candidate states against fixed references*.

3 Topic 10: What Factors Are and Why We Need Them (Section IV-C, Figure 3)

3.1 State Prediction Versus Factor Constraint

The IMU can propagate the previous state to create an initial guess for the next one. Yet a factor does not merely store that guess: it defines a residual measuring whether two *candidate* states

agree with integrated IMU measurements.

Key Idea

Prediction: "Using x_1 and new IMU data, I expect x_2."
IMU factor: "Whatever values the optimizer gives x_1 and x_2, their relative motion should agree with those IMU data."
Optimization: "Find the values best satisfying IMU, radar, barometer and other constraints together."

3.2 The Five Factor Types

Factor	Measurements/information used	What it constrains
Prior r_P	Startup reference values	Initial state x_0
IMU r_I	Accelerometer and gyroscope samples preintegrated between radar frames	Relative pose, velocity and bias consistency of x_i, x_{i+1}
Radar Doppler r_D	Each radar detection's bearing and radial Doppler	State at corresponding radar frame x_i
Barometer r_B	Pressure converted to relative altitude	Vertical position at a state x_i
Zero-velocity track r_T	Repeated matching static zero-Doppler radar reflections and range/bearing	Drone state x_i and tracked target position t^j

A radar frame may have 2, 1, or 4 detections; these produce 2, 1, or 4 Doppler factors respectively. Figure 3 in the paper depicts this variable count and one tracked landmark. The zero-velocity feature is not general landmark mapping throughout flight; the paper uses repeatable static detections around standstill.

3.3 IMU Preintegration Factor (Eq. 15)

Fast-rate IMU samples accumulate between consecutive radar frames. Rather than inserting an optimization factor for each IMU sample, preintegration summarizes relative rotation, position, and velocity changes in one factor, with bias residuals:

$$\boxed{r_I^{i,i+1} = \begin{bmatrix} r_{\Delta R}^{i,i+1} \\ r_{\Delta t}^{i,i+1} \\ r_{\Delta v}^{i,i+1} \\ r_{\Delta b_g}^{i,i+1} \\ r_{\Delta b_a}^{i,i+1} \end{bmatrix}}. \tag{12}$$

Illustrative Example (Not Experimental Data)

Assume a one-dimensional simplified motion model with $v_1 = 2$ m/s and the IMU indicating $\Delta v = 1$ m/s during t_1 to t_2. A candidate $v_2 = 2.8$ m/s gives $r_{\Delta v} \approx (2.8 - 2) - 1 = -0.2$ m/s. The true IMU factor additionally accounts for orientation, gravity, biases, and correlated uncertainty. The factor is connected to *both* x_1 and x_2.

3.4 Bearing Doppler Radar Factor (Eqs. 8–10)

For detection m at time i, the radar factor compares measured radial Doppler $v_T^{i,m}$ with the projection of the predicted radar velocity along measured unit bearing ${}^R e_T^{i,m}$:

$$r_D^{i,m} = -\left({}^R v_{IR}^i\right)^\mathsf{T} {}^R e_T^{i,m} - v_T^{i,m}, \tag{13}$$

$${}^I v_{IR}^i = {}^I v_{IB}^i + R_{IB}^i \left(\left({}^B \omega_{IB}^i - {}^B b_g^i\right) \times {}^B t_{BR}\right). \tag{14}$$

The second expression accounts for the radar being displaced from the IMU (the lever arm). Radar orientation calibration is used to transform velocity into radar coordinates. Each detection supplies a scalar radial constraint, *not* a full 3D velocity measurement by itself.

Illustrative Example (Not Experimental Data)

If the predicted Doppler is -1.0 m/s and the measured Doppler is -0.9 m/s, $r_D = -1.0 - (-0.9) = -0.1$ m/s. A moving cyclist may produce a much larger inconsistent residual; Welsch robust weighting limits its influence.

3.5 Barometer Factor (Eqs. 11–12)

Pressure p_i is converted into a barometric altitude z_{p_i} using the atmosphere model in Eq. (12). The paper uses a fixed initial offset z_{p_0}:

$$\boxed{r_B^i = \left({}^I e_z\right)^\mathsf{T} {}^I t_{IB}^i + z_{p_0} - z_{p_i}.} \tag{15}$$

This constrains the estimated *height*, not horizontal position. Downwash and ground effects can disturb pressure, motivating a Fair robust loss.

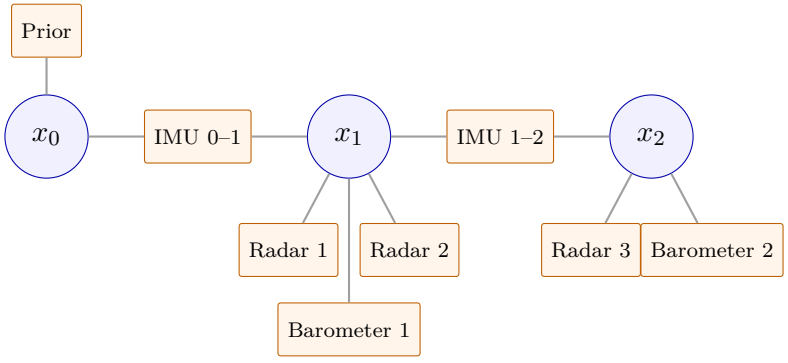
Illustrative Example (Not Experimental Data)

For an already offset-corrected altitude comparison, if the estimate is 0.5 m and the barometer-derived height is 0.4 m, the simplified residual is 0.1 m. The paper's residual explicitly includes the initial offset in Eq. (15).

3.6 Zero-Velocity Detection Tracking (Eqs. 13–14)

At standstill, CFAR may return only one or two radar targets, insufficient to constrain a general 3D velocity from Doppler alone. Identical repeatable zero-Doppler reflections can be associated across frames. A stationary target's inertial position t^j is estimated jointly with drone states. Each track factor compares that target's position predicted in the radar frame to its measured radar-frame detection position. This constrains both the target and the drone's pose at successive observations. The paper requires repeatable stationary detections; otherwise this aid is unavailable.

3.7 A Factor Graph with Three States



Schematic for learning: the number of radar factors can vary by frame; optional zero-velocity tracking factors connect to extra target-position nodes. The graph encodes dependencies; it does not itself perform numerical minimization.

4 Topic 11: Optimization, Uncertainty and Robust Loss (Section IV-C)

4.1 Why Not Simply Trust IMU Prediction?

Integrated IMU motion provides a useful *initial guess*, but its errors accumulate. Radar contributes velocity information and barometry height information. The optimizer finds states best satisfying all factor constraints, rather than blindly selecting one sensor's answer.

4.2 Maximum A Posteriori Estimation (Eq. 6)

With X_k the collection of states, T_k the collection of tracked target locations, and Z_k the sensor measurements, the paper writes

$$\boxed{(X_k^{\mathrm{MAP}}, T_k^{\mathrm{MAP}}) = \underset{X_k, T_k}{\operatorname{argmax}}\ p(X_k, T_k \mid Z_k).} \tag{16}$$

That means: find the most probable states and tracked targets given the measurements and assumptions. Minimizing a sum of weighted residual costs yields an equivalent optimization problem under the chosen noise models.

4.3 Why We Do Not Add Raw Errors

A radar Doppler residual has units of m/s; a barometer residual has units of m; an IMU residual combines rotation, velocity, position, and bias terms. Simply adding numerical residuals is not meaningful. Instead each residual is weighted relative to its *uncertainty*. For a scalar measurement with standard deviation σ:

$$\boxed{r_{\mathrm{normalized}} = \frac{r}{\sigma}, \qquad J_{\mathrm{quadratic}} = \left(\frac{r}{\sigma}\right)^2.} \tag{17}$$

Normalization means expressing an error in units of its expected standard deviation: $r/\sigma = 2$ means the residual is two noise standard deviations, regardless of the original physical units. Normalizing does not repair a bad sensor reading.

Illustrative Example (Not Experimental Data)

Radar residual 0.1 m/s with $\sigma_D = 0.05$ m/s gives normalized residual 2 and ordinary quadratic cost 4. Barometer residual 0.4 m with $\sigma_B = 0.2$ m gives normalized residual 2 and quadratic cost 4. These *equally normalized* residuals are not proof that the sensors are equally accurate. Actual BRIO Doppler and barometer terms also include robust weighting.

4.4 Standard Deviation, Variance and Covariance

Standard deviation σ describes expected measurement spread. Variance is σ^2. A multidimensional residual requires a covariance matrix Σ, describing error variance and possibly correlation across components.

Illustrative Example (Not Experimental Data)

For three *uncorrelated* errors with standard deviations $0.1, 0.2, 0.3$ in compatible respective units:

$$\Sigma = \begin{bmatrix} 0.01 & 0 & 0 \\ 0 & 0.04 & 0 \\ 0 & 0 & 0.09 \end{bmatrix}, \quad r = \begin{bmatrix} 0.2 \\ 0.1 \\ 0.3 \end{bmatrix}.$$

The off-diagonal entries are zero only because independence is assumed for this illustration; a general IMU preintegration covariance need not be diagonal.

4.5 Squared Mahalanobis Distance

The paper defines

$$\boxed{\|r\|_\Sigma^2 = r^\mathsf{T}\Sigma^{-1}r.} \tag{18}$$

The inverse covariance gives higher penalties for errors in directions with smaller uncertainty. In one dimension, $\Sigma = \sigma^2$ and Eq. (18) becomes r^2/σ^2.

Illustrative Example (Not Experimental Data)

For the diagonal 3D example above,

$$\|r\|_\Sigma^2 = \frac{0.2^2}{0.01} + \frac{0.1^2}{0.04} + \frac{0.3^2}{0.09} = 4 + 0.25 + 1 = \boxed{5.25}.$$

Thus multidimensional errors can be combined into one uncertainty-weighted scalar cost.

4.6 Why Lower Uncertainty Is a Stronger Constraint

For the *same* scalar residual $r = 0.2$ and equal units:

$$\sigma_A = 0.1: \quad J_A = \frac{0.2^2}{0.1^2} = 4, \qquad \sigma_B = 0.5: \quad J_B = \frac{0.2^2}{0.5^2} = 0.16.$$

A low-uncertainty sensor imposes a larger penalty for a given disagreement. This is why smaller σ normally means a stronger constraint, *provided the noise model is realistic.*

4.7 Complete Two-Sensor Velocity Example

Suppose two hypothetical sensors measure the *same scalar velocity*:

$$z_A = 1.0 \text{ m/s}, \ \sigma_A = 0.1 \text{ m/s}, \qquad z_B = 1.5 \text{ m/s}, \ \sigma_B = 0.5 \text{ m/s}.$$

For candidate velocity v, the ordinary quadratic objective is

$$J(v) = \frac{(v-1.0)^2}{0.1^2} + \frac{(v-1.5)^2}{0.5^2} \tag{19}$$

$$= 100(v-1)^2 + 4(v-1.5)^2. \tag{20}$$

At $v = 1.2$ m/s, residuals are 0.2 and -0.3 m/s; the costs are 4 and 0.36, so $J = 4.36$. To find the best compromise, solve $dJ/dv = 0$:

$$200(v-1) + 8(v-1.5) = 0, \tag{21}$$

$$208v = 212, \tag{22}$$

$$\boxed{v^* = \frac{106}{104} \approx 1.01923 \text{ m/s}.} \tag{23}$$

The optimum is much nearer 1.0 m/s because Sensor A's noise is smaller. This example illustrates weighting only; actual radar detections are directional radial constraints rather than full forward-velocity measurements.

4.8 Why Robust Loss Is Needed

Ordinary quadratic cost grows as r^2. A moving cyclist or ghost radar target can have a very large Doppler residual and dominate the fit; barometric pressure can jump near the ground due to propeller effects. A *robust loss* reduces the influence of such unusually large errors. The paper compares alternative losses and chooses:

Factor	Treatment in the paper	Motivation
Prior	Covariance-weighted quadratic	Anchor initialization
IMU	Covariance-weighted quadratic	Relative-motion consistency
Zero-velocity tracking	Covariance-weighted quadratic	Repeated stationary-detection consistency
Radar Doppler	Welsch robust loss	Moving objects / ghost Doppler detections
Barometer	Fair robust loss	Pressure disturbances from aerodynamic effects

Welsch has a bounded cost whose influence falls toward zero for very large errors (a redescending loss). **Fair** grows more gently than quadratic at large residuals but continues to grow. Robust loss downweights a measurement; it need not explicitly delete it. Normalization answers *how many noise standard deviations away*; robust loss answers *how strongly to penalize a very large discrepancy.*

4.9 The Complete Objective: Structure of Eq. (7)

For readability, write the paper's objective as the sum of categories:

$$\boxed{\min_{X_k,T_k} J(X_k,T_k), \qquad J = J_P + J_I + J_T + J_D + J_B.} \tag{24}$$

More explicitly, following the structure of Eq. (7),

$$J_P = \left\| r_P^0 \right\|_{\Sigma_P}^2 , \tag{25}$$

$$J_I = \sum_{(i,i+1)} \left\| r_I^{i,i+1} \right\|_{\Sigma_I}^2 , \tag{26}$$

$$J_T = \sum_j \sum_{i \in K_j} \left\| r_T^{i,j} \right\|_{\Sigma_T}^2 , \tag{27}$$

$$J_D = \sum_i \sum_{m \in \mathcal{R}_i} \rho_{\mathrm{W}} \left(\frac{|r_D^{i,m}|}{\sigma_D} \right) \left\| r_D^{i,m} \right\|_{\Sigma_D}^2 , \tag{28}$$

$$J_B = \sum_i \rho_{\mathrm{F}} \left(\frac{|r_B^i|}{\sigma_B} \right) \left\| r_B^i \right\|_{\Sigma_B}^2 . \tag{29}$$

Notation note: The paper writes ρ as a robust weighting applied to covariance-weighted squared errors, not simply $\rho(r/\sigma)$ as a standalone cost. This notation follows that presentation; robust optimization software may implement the same broad idea via a robust cost/kernel and an associated iteratively changing weight. $\mathcal{R}_i$ is the set of detections at radar frame i, while K_j indexes the frames observing track j. The covariance arguments in software may vary with time and factor even if the paper suppresses some indices.

5 What the Optimizer Actually Does

Given current guesses for state values, the optimizer:

1. Predicts the measurement expected by each factor from those guesses.
2. Calculates residuals against the stored measurements.
3. Uses covariance to scale each residual; applies robust weighting where specified.
4. Sums the factor costs into J.
5. Computes corrections to state variables (including rotations on SO(3)) that reduce J.
6. Updates the estimates and repeats as required for convergence.

The optimizer does *not* independently force every factor's residual to zero. It finds a compromise that minimizes their total weighted cost. Previous states can also be revised as more constraints arrive.

6 GTSAM, iSAM2, and Online Updates

GTSAM is the software library used to represent state variables, priors, IMU factors, custom radar/barometer factors, noise models, and the nonlinear factor graph. **iSAM2** is an incremental nonlinear solver used to update the estimate efficiently as new states and factors arrive, reusing previous calculations.

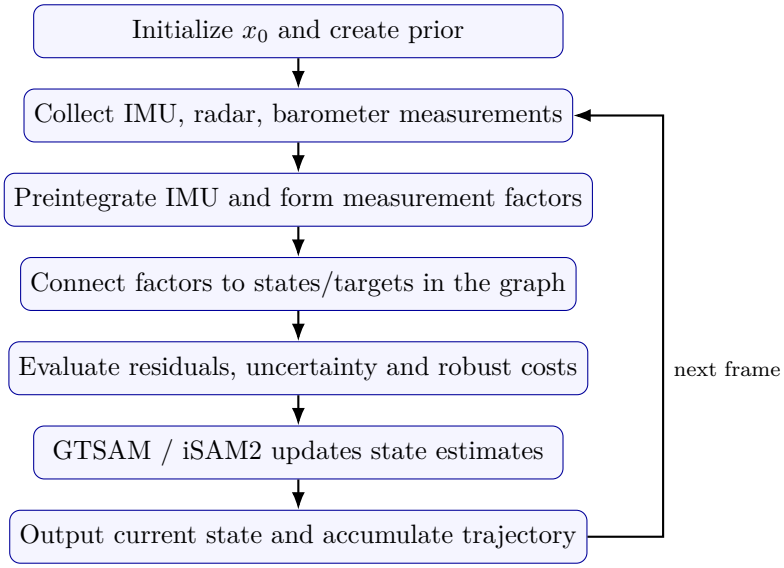
What the Paper States

Section V reports optimization implemented in GTSAM with iSAM2, using two threads and a 10-second data window. An optimization thread updates the MAP solution as radar frames arrive, typically within about 40 ms; a navigation thread predicts the latest state at IMU rate for the flight controller. The reported experimental rates include IMU 400 Hz, radar approximately 8 Hz, and barometer 50 Hz.

A *prediction thread* gives high-rate motion estimates between slower radar-linked optimization updates. It is not a second independent factor-graph solution.

7 A Complete Three-State Timeline

1. **At t_0:** Keep the drone still, initialize x_0, average initial gyro samples for bias, obtain orientation reference, create the prior factor.
2. **Between t_0 and t_1:** Collect many IMU measurements; preintegrate them. IMU propagation may provide a first guess for x_1.
3. **At radar frame t_1:** Create x_1. Add IMU factor (x_0, x_1), one Doppler factor per radar detection, and a barometer factor using synchronized pressure. Add zero-velocity track factors when appropriate.
4. **Optimize:** Evaluate all factor residuals and minimize their combined uncertainty-weighted robust objective, updating estimates such as x_0 and x_1.
5. **Between t_1 and t_2:** Gather/preintegrate new IMU data; predict the next state.
6. **At radar frame t_2:** Create x_2 and IMU factor (x_1, x_2); add new radar/barometer/optional tracking factors and update the graph using iSAM2.
7. **Output:** Extract the optimized positions, orientations, velocities, and biases. Positions across time form the estimated trajectory.



8 What We Will Reproduce in Code

1. Parse and synchronize IMU, radar and (when available) barometer measurements.
2. Define the 15-DoF drone state and radar-to-IMU extrinsic calibration.
3. Initialize x_0, create prior references and noise model, and estimate startup gyroscope bias.
4. Preintegrate IMU data between radar frames; add IMU factors.
5. For each detection add a bearing Doppler factor with a Welsch robust model.
6. Convert barometric pressure into a relative altitude constraint with Fair robust treatment.
7. Implement repeatable zero-Doppler target association and track-position variables for standstill factors.
8. Construct and update the graph in GTSAM/iSAM2; extract states and trajectories.
9. Assess drift and trajectory consistency against the paper's reported experiments.

Key Idea

Dataset limitation: The paper states that the ColoRadar benchmark does *not* supply barometer measurements, so its Section VI benchmark omits the barometer factor. Full BRIO reproduction therefore requires recordings with usable IMU, radar *and* barometer data; ColoRadar can support a radar-inertial subset.

9 Short Answers to Recurring Questions

"Why factors if IMU can calculate the next state?"
IMU-only propagation drifts. Factors allow the solver to reconcile that propagation with radar and altitude information in one optimized solution.

"Is the IMU factor calculated at x_1 or x_2?"
The preintegrated factor spans the time interval from t_1 to t_2 and depends on *both* x_1 and x_2. Its residual is evaluated using both candidate states.

"Do measurements go to optimizer before graph?"
No. Measurements are incorporated into factors, factors are connected to unknown states in the graph, and the optimizer evaluates the graph's residual functions.

"Why compute prior residuals after initialization?"
Initialization provides fixed reference values. Prior residuals compare candidate estimates against them during optimization.

"Does normalize mean remove noise?"
No. It expresses a residual in units of expected noise: r/σ.

"What if sensor errors have different units?"
Use each measurement's covariance to form unitless weighted costs before summing them.

"Does each radar detection give full 3D velocity?"
No. Each Doppler detection gives one radial component. Multiple nondegenerate bearings constrain 3D velocity.

"Which losses does the paper use?"
Quadratic for prior, IMU and track; Welsch for Doppler; Fair for barometry.

Reference

R. Girod, M. Hauswirth, P. Pfreundschuh, M. Biasio, and R. Siegwart, "A Robust Baro-Radar-Inertial Odometry M-Estimator for Multicopter Navigation in Cities and Forests," arXiv:2408.05764v1, 2024. Main sources for this document: Section IV-C (Eqs. 6–7, Fig. 3), Section IV-D (Eqs. 8–10), Section IV-E (Eqs. 11–12), Section IV-F (Eqs. 13–14), Section IV-G (Eq. 15), Section IV-H (Eqs. 16–21), and Section V (implementation). Code and data: https://github.com/ethz-asl/rio.